



Intro to R



A crash course to basic R

Instructor: Sahifa Siddiqua



Objective

These slides will help you do the following things:

- Download R and RStudio onto your laptops
- Familiarize yourselves with the interface of RStudio
- Learn the use of the different tabs in RStudio

Please make sure you bring your laptops to class and have R and RStudio installed by the first day of classes.



Software necessary for R

- R: A freely available language and environment for statistical computing and graphics, which provides a wide variety of statistical and graphical techniques: linear and nonlinear modelling, statistical tests, time series analysis, classification, clustering, etc. Everything you need to know about R is [here](#).
- Rstudio: A free integrated development environment (IDE) designed to support multiple languages, including both R and Python. It's a user-friendly interface that makes documenting and working with R a lot easier. Everything you need to know about RStudio is [here](#).



Logo and User Interfaces

R:



Logo

```
R Console

R version 4.4.1 (2024-06-14) -- "Race for Your Life"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: x86_64-apple-darwin20

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[R.app GUI 1.80 (8416) x86_64-apple-darwin20]

[Workspace restored from /Users/s.sahifa98/.RData]
[History restored from /Users/s.sahifa98/.Rapp.history]

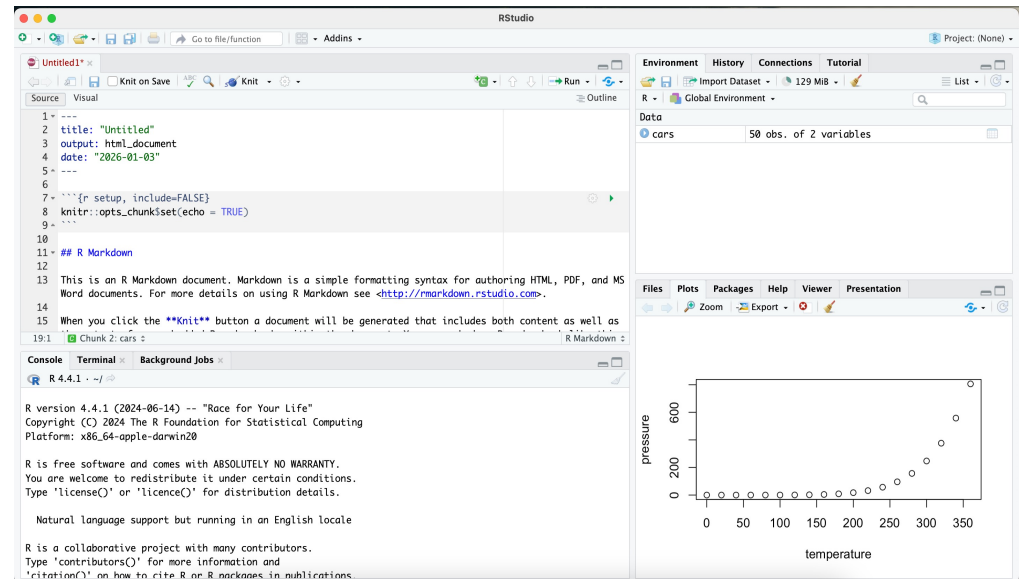
>
```

User Interface

RStudio:



Logo



User Interface



Installing R and RStudio



On Your Individual Device



Installing R and RStudio

- Leave directories and settings as **default** unless you know what you're doing!
- <https://posit.co/download/rstudio-desktop/>
- First, click “Download and Install R”. Click “Download R for macOS” or “Download R for Windows” depending on what type of computer you have, in the top box.
 - MacOS: Click the .pkg file that applies to your computer
 - Windows: Click “Install R for the first time”, and then “Download R-4.3.2 for Windows”
- Then, back at the above link, click “Download RStudio Desktop”, which will just give you a file.



Steps to download R and R-studio

- Web browser -> search “rstudio desktop” -> select “[Rstudio desktop](#)” by Posit (the developer of RStudio)
- Or go to the Posit website “<https://posit.co/download/rstudio-desktop/>” (official website for RStudio)

1: Install R

RStudio requires R 3.6.0+. Choose a version of R that matches your computer's operating system.

R is not a Posit product. By clicking on the link below to download and install R, you are leaving the Posit website. Posit disclaims any obligations and all liability with respect to R and the R website.

DOWNLOAD AND INSTALL R

2: Install RStudio

DOWNLOAD RSTUDIO DESKTOP FOR MACOS 12+

This version of RStudio is only supported on macOS 12 and higher. For earlier macOS environments, please [download a previous version](#).

Size: 664.40 MB | [SHA-256: D0DDD395](#) | Version: 2024.04.2+764
| Released: 2024-06-10



Steps to download R

- Takes you to the official website for R (“<https://cran.rstudio.com>”)

The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- [Download R for Linux](#) ([Debian](#), [Fedora/Redhat](#), [Ubuntu](#))
- [Download R for macOS](#)
- [Download R for Windows](#)

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.



Latest version of R is R-4.4.1

R for Windows

Subdirectories:

[base](#)

Binaries for base distribution. This is what you want to **install R for the first time.**

[contrib](#)

Binaries of contributed CRAN packages (for R $\geq$ 4.0.x).

[oldcontrib](#)

Binaries of contributed CRAN packages for outdated versions of R (for R $<$ 4.0.x).

[Rtools](#)

Tools to build R and R packages. This is what you want to build your own packages on Windows, or to build R itself.

R for macOS

This directory contains binaries for the base distribution and of R and packages to run on macOS. R and package binaries for R versions older than 4.0.0 are only available from the [CRAN archive](#) so users of such versions should adjust the CRAN mirror setting (<https://cran-archive.r-project.org>) accordingly.

Note: Although we take precautions when assembling binaries, please use the normal precautions with downloaded executables.

R 4.4.1 "Race for Your Life" released on 2024/06/14

Please check the integrity of the downloaded package by checking the signature:

```
pkgutil --check-signature R-4.4.1-arm64.pkg
```

in the *Terminal* application. If Apple tools are not available you can check the SHA1 checksum of the downloaded image:

```
openssl sha1 R-4.4.1-arm64.pkg
```

Latest release:

For Apple silicon (M1-3) Macs:

[R-4.4.1-arm64.pkg](#)

SHA1-hash: 616560b17092b6dd8b814d9cd92d098e5220f830
(ca. 94MB, notarized and signed)

For older Intel Macs:

[R-4.4.1-x86_64.pkg](#)

SHA1-hash: e66eb09244121d7db78fb41d3c06a78796c93b5
(ca. 96MB, notarized and signed)

R 4.4.1 binary for macOS 11 (**Big Sur**) and higher, signed and notarized packages.

Contains R 4.4.1 framework, R.app GUI 1.80, Tcl/Tk 8.6.12 X11 libraries and Texinfo 6.8. The latter two components are optional and can be omitted when choosing "custom install", they are only needed if you want to use the `tcltk` R package or build package documentation from sources.

macOS Ventura users: there is a known bug in Ventura preventing installations from some locations without a prompt. If the installation fails, move the downloaded file away from the *Downloads* folder (e.g., to your home or Desktop).

Note: the use of X11 (including `tcltk`) requires [XQuartz](#) (version 2.8.5 or later). Always re-install XQuartz when upgrading your macOS to a new major version.

This release uses Xcode 14.2/14.3 and GNU Fortran 12.2. If you wish to compile R packages which contain Fortran code, you may need to download the corresponding GNU Fortran compiler from <https://mac.R-project.org/tools>. Any external libraries and tools are expected to live in `/opt/R/arm64` (Apple silicon) or `/opt/R/x86_64` (Intel).



- Go back to the Rstudio website after you download and install R.

1: Install R

RStudio requires R 3.6.0+. Choose a version of R that matches your computer's operating system.

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DOWNLOAD AND INSTALL R

2: Install RStudio

DOWNLOAD RSTUDIO DESKTOP FOR MACOS 12+

This version of RStudio is only supported on macOS 12 and higher. For earlier macOS environments, please [download a previous version](#).

Size: 664.40 MB | [SHA-256: D0DDD395](#) | Version: 2024.04.2+764
| Released: 2024-06-10

OS	Download	Size	SHA-256
Windows 10/11	RSTUDIO-2024.04.2-764.EXE ⬇	262.79 MB	09E1E38A



RStudio





User Interface

We will use the image below to understand the 4 primary panes/windows of Rstudio. Each Pane will be discussed in details in the following slides

The screenshot displays the RStudio interface with four main panes highlighted by red boxes and labeled with red text:

- Pane 1 (Source):** Shows the R Markdown document editor. The code includes a YAML header, R setup code, and R Markdown content. The text "Pane 1" is overlaid in red.
- Pane 2 (Console):** Shows the R version (4.4.1) and copyright information. The text "Pane 2" is overlaid in red.
- Pane 3 (Environment):** Shows the Global Environment with a data frame named 'cars' containing 50 observations of 2 variables. The text "Pane 3" is overlaid in red.
- Pane 4 (Plots):** Shows a scatter plot of 'pressure' vs 'temperature' from the 'cars' dataset. The text "Pane 4" is overlaid in red.

The R Markdown code in Pane 1 is as follows:

```
1 ---
2 title: "Untitled"
3 output: html_document
4 date: "2026-01-03"
5 ---
6
7 {r setup, include=FALSE}
8 knitr::opts_chunk$set(echo = TRUE)
9
10
11 ## R Markdown
12
13 This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS
14 Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.
15
16 When you click the **Knit** button a document will be generated that includes both content as well as
```



User Interface

When you open RStudio for the first time it will only not show Pane 1. This is because Pane 1 is primarily for displaying R format documents. So let's understand what you will see for the first time. NOTE: All pane names are the same as the image in previous slide and is missing Pane 1.

The screenshot displays the RStudio interface with three visible panes:

- Pane 2 (Console):** Shows the R version 4.4.1 (2024-06-14) and the "Race for Your Life" project. It includes the R license text and instructions for using R, such as `demo()`, `help()`, `help.start()`, and `q()`. The workspace is loaded from `~/RData`.
- Pane 3 (Environment):** Displays the Global Environment, which is currently empty.
- Pane 4 (Files):** Shows the file explorer with a list of files and folders, including `.RData`, `.Rhistory`, `Applications`, `bi`, `copula-2.pdf`, `Desktop`, `Documents`, `download.csv`, `Downloads`, `GlobalProtect_Logs.tgz`, and `HMMna`.



We will start by understanding other Panes and then at last look at Pane 1.

Pane 2: Console Pane

- By default, the console pane is the bottom left pane. The **Console** tab provides an interactive area for executing code. By default, it is tied to R and looks similar to the R interface. This is the brain of RStudio and uses the R that you installed. Without installing R, you can't use the console.
- You can use the console to write your code directly. But it doesn't save them in a file for future use. To save your code in a file, use an R format file in the **SOURCE** tab.
- The Console pane also includes an integrated **Terminal** tab for executing system commands, adding or removing additional integrated terminals, and controlling the current selected terminal.
- The **Background Jobs** tab provides the ability to send long-running R scripts to local and remote background jobs. This functionality can dramatically improve the productivity of data scientists and analysts using R, as they can continue working in RStudio while jobs run in the background.



Pane 3: Environment Pane

- By default, the Environment pane is located in the top-right and includes the Environment, History, Connections, Build, and Version Control System (VCS) tabs.
- The **Environment tab** displays currently saved R objects. For example, the following code would add 'x' with value 1 to the environments pane for the duration of the current session.

```
| > x = 1 |
```

- The environment menu bar provides loading or saving of R Workspaces, interactive import of datasets from text files, Excel, or SPSS/SAS/Stata. It also displays the memory currently used by the active R session, along with a broom icon to remove all current environment objects.
- The **History tab** displays the commands that were executed in the current session, along with search. There are buttons for loading/saving the command history to a file, as well as sending the selected command into the console or inserting it into the current working document. There is a delete button to delete the selected history, or a broom button to remove all history for the current session.

These are the only two relevant tabs for the course. If you are interested in other tabs, see [this](#). An example of the Environment tab from the Environment Pane is shown in the next slide.



The Environment tab is one of the most useful tabs.

The screenshot shows the RStudio Environment tab. At the top, there are tabs for 'Environment', 'History', 'Connections', and 'Tutorial'. Below these are icons for file operations (Import, Save, etc.) and a memory usage indicator showing '221 MiB'. The main area is divided into two sections: 'Data' and 'Values'. The 'Data' section contains a table with two rows: 'data' (50 obs. of 2 variables) and 'model' (List of 13). The 'Values' section contains a table with two rows: 'vector' (num [1:3] 1 2 3) and 'x' (1). Red boxes highlight the 'Data' and 'Values' sections. Red arrows point from the blue play icons in the 'Data' section to the right, and from the 'Values' section to the right.

Data	
data	50 obs. of 2 variables
model	List of 13

Values	
vector	num [1:3] 1 2 3
x	1

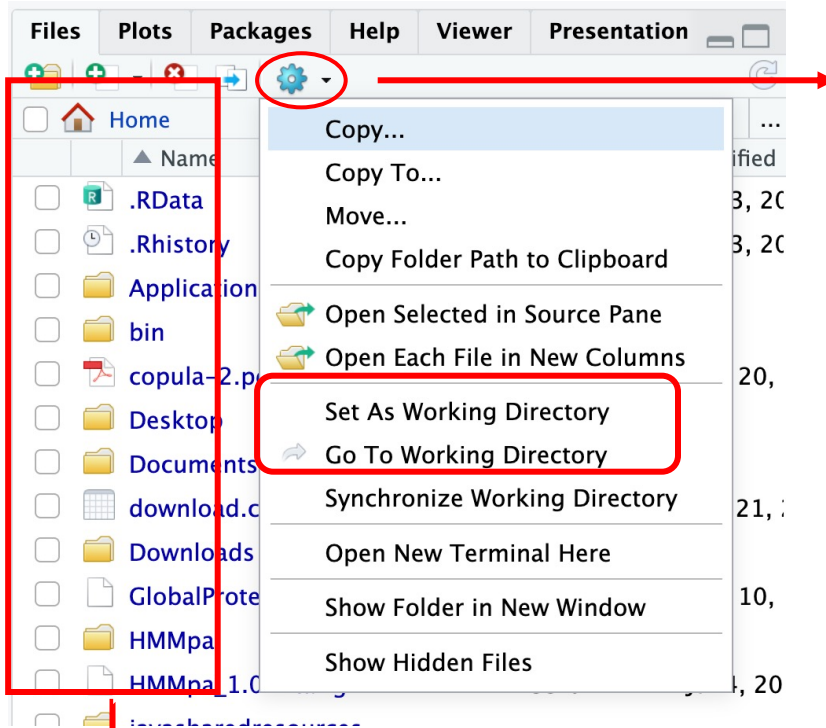
All data and big objects are saved like this, where clicking on the blue circles will display more info about the object

Other vector-related R objects are saved here



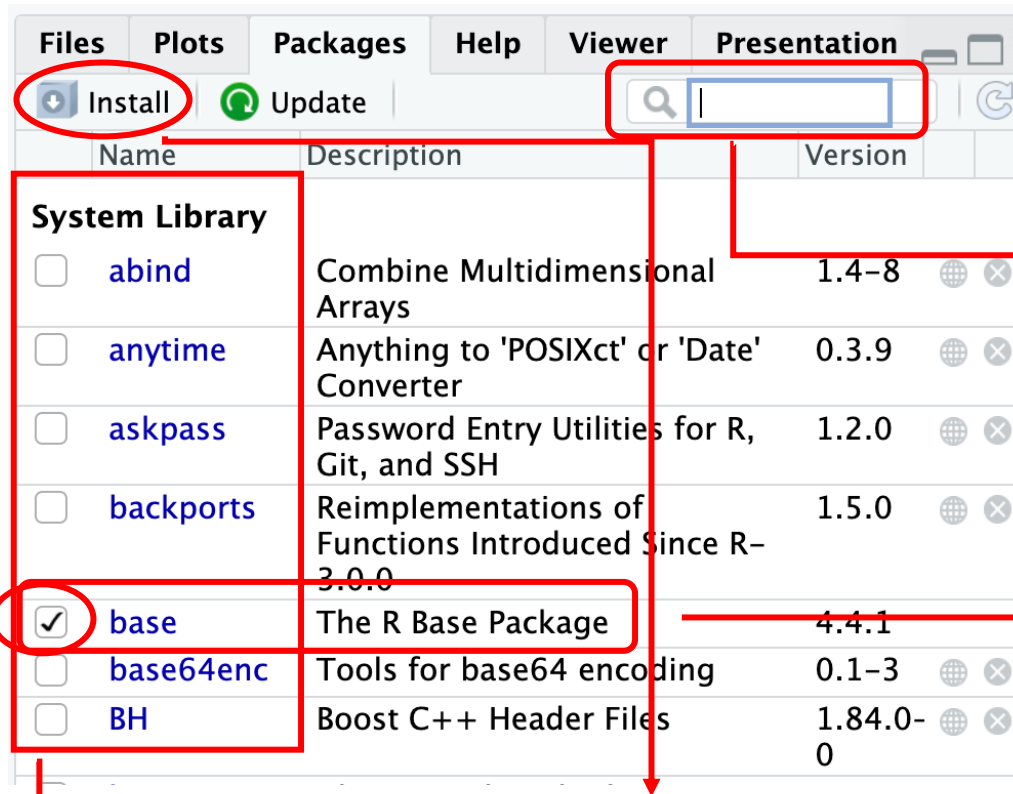
Pane 4: Output Pane

- By default, the Output pane is the bottom-right pane and displays various outputs such as plots, HTML content, or on-disk files. It contains the Files, Plots, R Packages, Help, Tutorial, Viewer, and Presentation tabs.
- **Files tab** provides an interactive exploration of the current R Project along with the entire directory of the computer. There are menu bar options for adding new folders and new blank files, as well as for deleting or renaming files, as well as a gear pane for additional functions. One of the main uses of the gear pane is to set the working directory. We will go over that later in the slides.
- **Plots tab** will display static images generated by code run in the console or Rscript file until the session is restarted.
- **Packages tab** allows for viewing all installed R packages, and has a search bar for searching the library of packages.
- **Help tab** is used to display package and function documentations. To get help on a specific R function, use the `?functionName` or `help(functionName)` or just type in the function's name in the search bar of the help tab.
- **Viewer tab** is used to display web content such as HTML files, Shiny apps, Quarto-generated web pages, or interactive HTMLwidgets like plotly graphs. All your knitted HW and other files will be displayed here. (More on knitting in later slides)



The widget in this course primarily sets the directory. We will go over this later.

Files tab shows all the folders and files on the computer. You can navigate these folder to find your desired files/folders here.



The search panel can be used to search for packages

Loaded packages are selected here to indicate that they are loaded for the current session

The default R package

R primarily comes with some packages which gets installed when your are downloading the software. But you can install new packages by running the code `install.packages("name of the package")` or by using the install button here.



The package where the function is defined

The screenshot shows the RStudio interface. At the top, there are tabs for Files, Plots, Packages, Help, Viewer, and Presentation. Below these is a search bar containing the text 'lm'. A red arrow points from the search bar to the right, towards the text 'Search bar can be used to search for functions, datasets, etc that come in a package'. Below the search bar, the 'R: Fitting Linear Models' dropdown is visible. A red arrow points from the 'lm(stats)' text in the dropdown to the left, towards the text 'The package where the function is defined'. The main content area displays the documentation for the 'lm' function, titled 'Fitting Linear Models'. This area is enclosed in a red rounded rectangle. A red arrow points from the bottom of this rectangle to the text 'Detailed explanation of the function, including usage, input variables, output variables, examples, etc'.

Files Plots Packages Help Viewer Presentation

R: Fitting Linear Models Find in Topic

lm(stats) R Documentation

Fitting Linear Models

Description

lm is used to fit linear models, including multivariate ones. It can be used to carry out regression, single stratum analysis of variance and analysis of covariance (although [aov](#) may provide a more convenient interface for these).

Usage

```
lm(formula, data, subset, weights, na.action,
  method = "qr", model = TRUE, x = FALSE, y = TRUE,
  singular.ok = TRUE, contrasts = NULL, offset)
```

Search bar can be used to search for functions, datasets, etc that come in a package

Detailed explanation of the function, including usage, input variables, output variables, examples, etc



Demo rmd file

Sahifa

2026-01-03

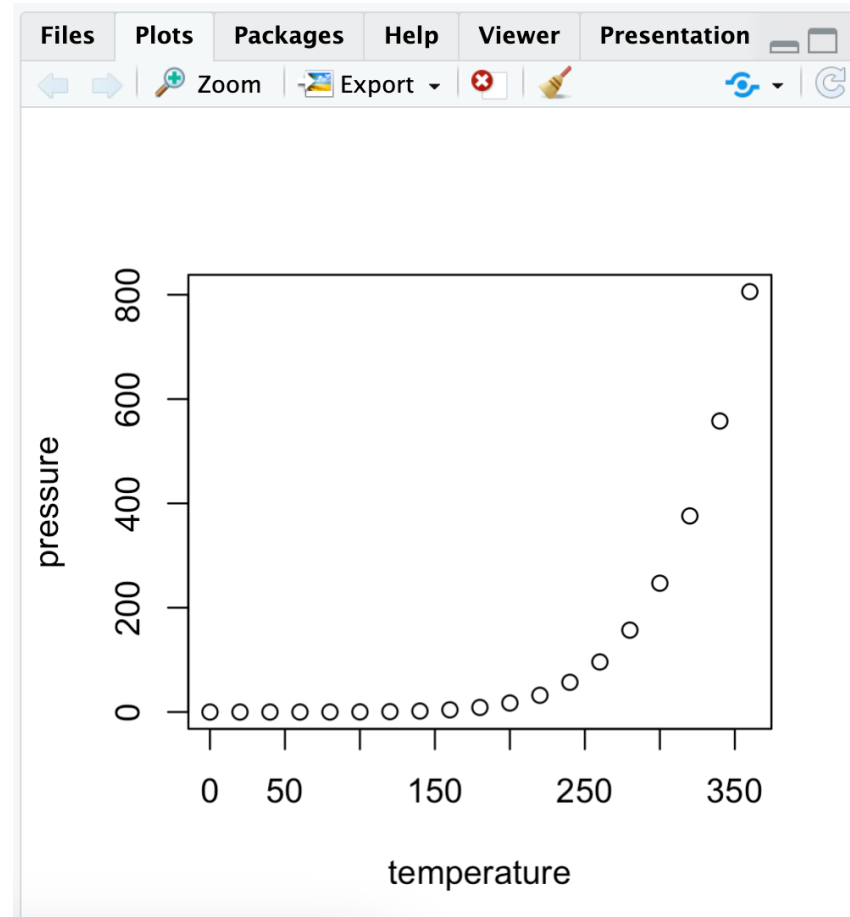
R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

The Plots tab shows the plots made in the console or R files that don't have an inline display, such as the .R file.

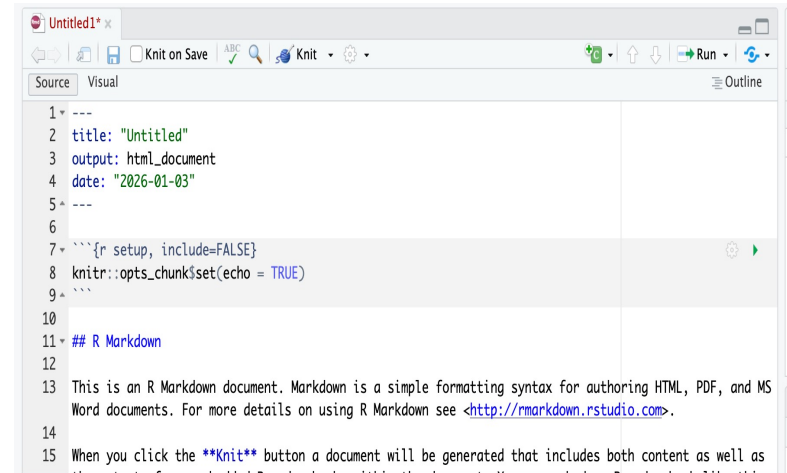
The Viewer tab shows the HTML file obtained from knitting an .Rmd





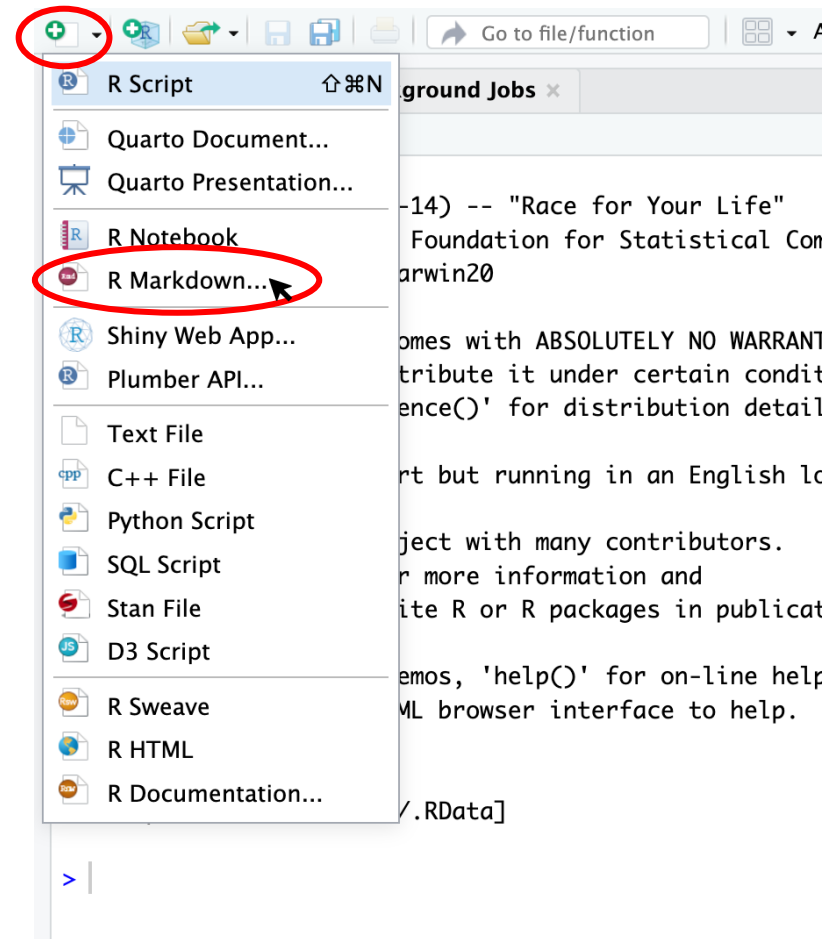
Pane 1: Source Pane

- The source pane allows users to view and edit various code-related files, such as .R, .rmd, .qmd, .py, .css, or general text files such as .txt or .md. By default it is the top-left panel and can be launched by opening any editable file in RStudio. Each additional file that is opened will be added as a new tab within the Source pane.
- For this course, we will primarily be working with an R Markdown file. The format of this file is .rmd and can only be opened in R. All your HWs, DIS and lecture notes will be in this format.

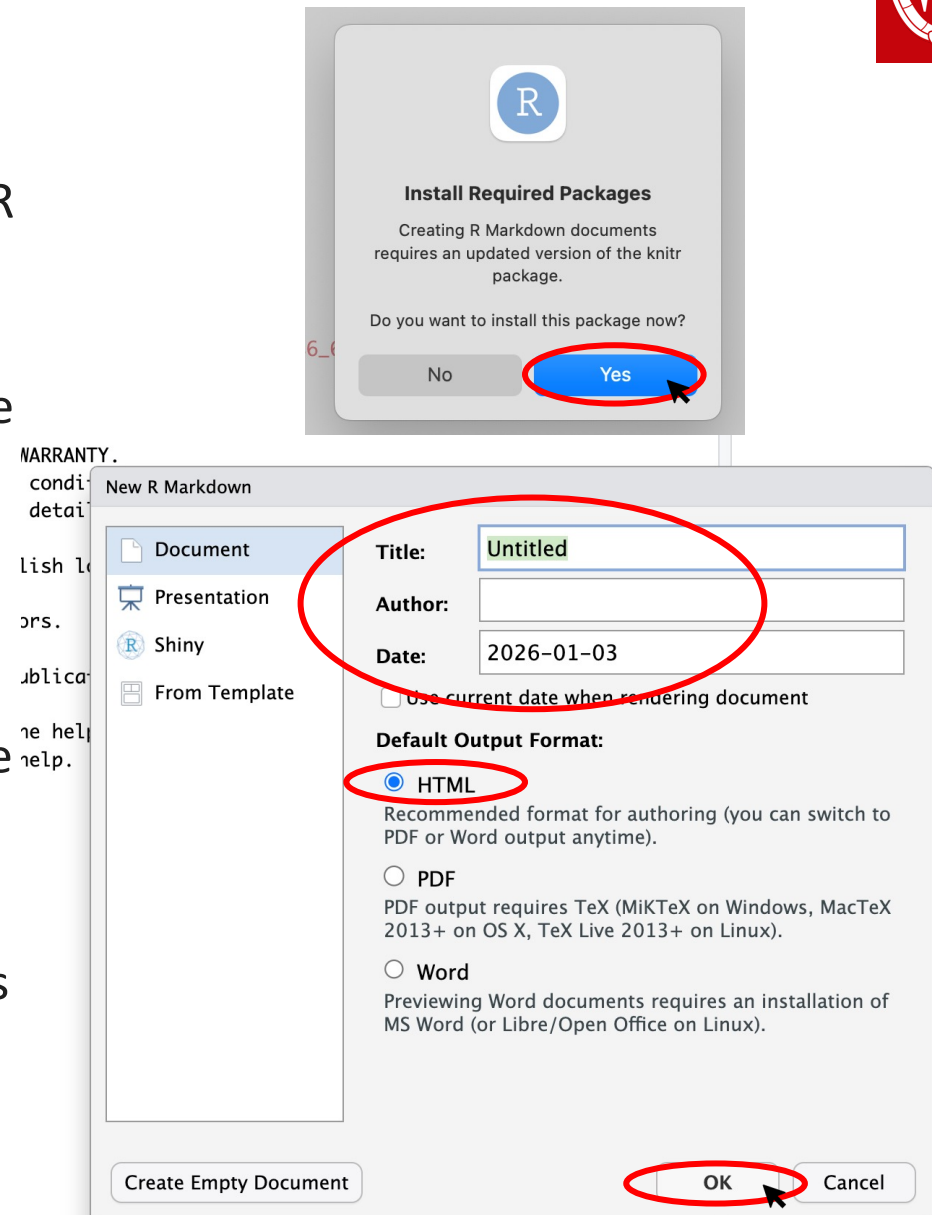


Creating an R format file in Source pane

1. When you open a new RStudio window, it will not have an open source window, since you haven't opened or created an R format file.
2. To create an R format file, such as an R Markdown file, click the green + sign in the top-left corner of your RStudio window.
3. A drop-down menu will appear, as shown in the image. Choose the option of R Markdown to open a new R Markdown file.
4. You can also create other R format files such as the R Script or Text File, to name a few.



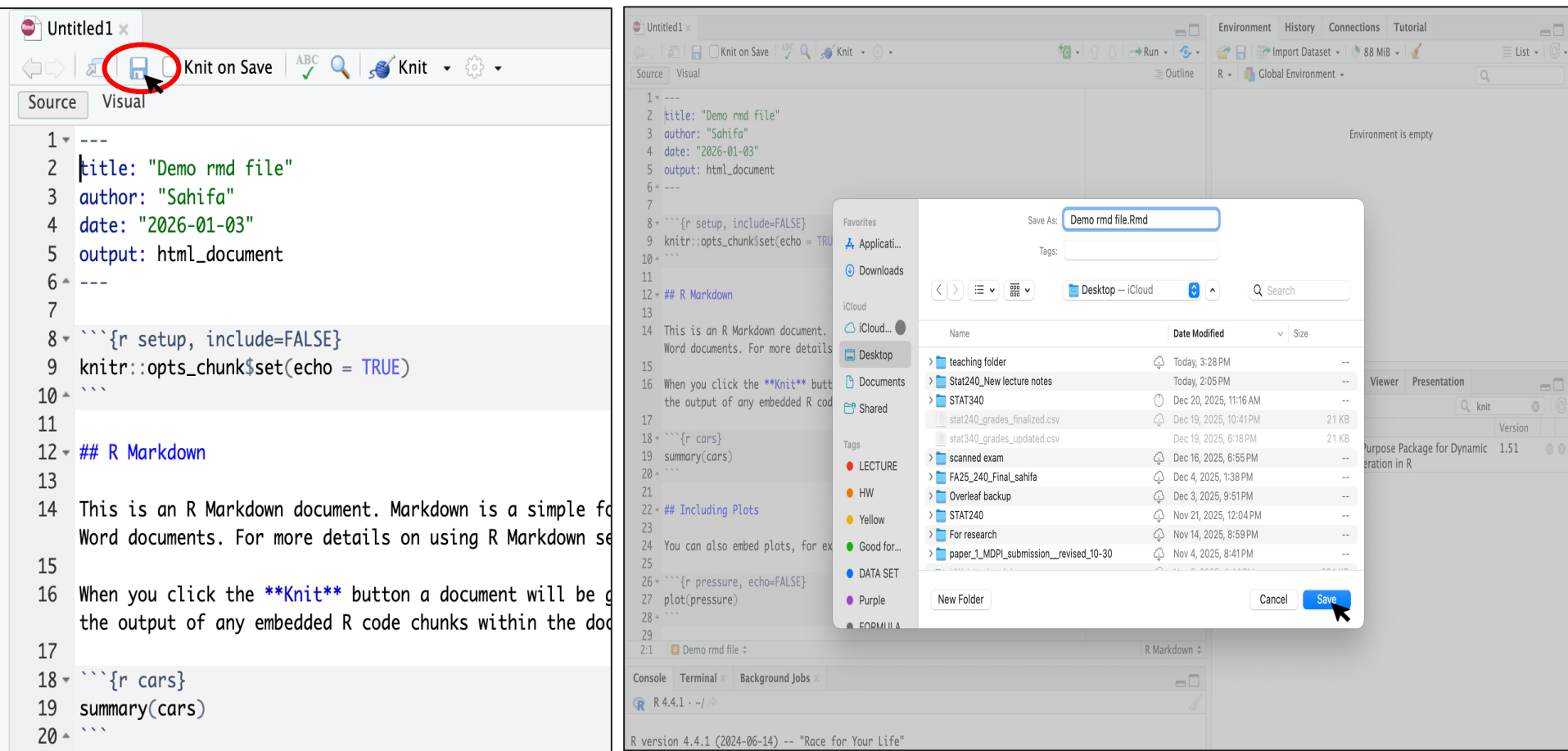
4. Once you select the R Markdown option, a prompt similar to the one shown in the image below will open. For first-time users, it will prompt you to download the necessary packages to create an R Markdown file. Just click Yes, and it will automatically start downloading the required packages. Once done, you will see the window as shown in the image below.
5. Here you can select the title, author, date, and format of the output document.
6. Once you click OK, a new .rmd file will be created.
7. In this course, we primarily want the output in HTML format. But you can also generate output files as a pdf or word. You might need additional packages to do so.





8. A new R Markdown file looks like the image below. You can see in the image below that the file's name in the top tab is "Untitled1". This is because the file has not been saved yet. To save the file, click on the  icon and select the location where you want to save it on your computer. Once saved, the file name will be displayed accordingly.

For the course, it's suggested that you save .Rmd files in the folders related to what you are working with. For example, if you are working on your HW1, save the .Rmd file in the HW -> HW1 folder of your STAT240 folder.



The image shows two screenshots of the RStudio interface. The left screenshot shows a new R Markdown file named "Untitled1" with the following content:

```
1 ---
2 title: "Demo rmd file"
3 author: "Sahifa"
4 date: "2026-01-03"
5 output: html_document
6 ---
7
8 ```{r setup, include=FALSE}
9 knitr::opts_chunk$set(echo = TRUE)
10 ```
11
12 ## R Markdown
13
14 This is an R Markdown document. Markdown is a simple formatting syntax for
15 Word documents. For more details on using R Markdown see
16
17 When you click the **Knit** button a document will be generated containing
18 the output of any embedded R code chunks within the document.
19
20 ```{r cars}
21 summary(cars)
22 ```
```

The right screenshot shows the same R Markdown file with a save dialog open. The dialog is titled "Save As:" and shows the file name "Demo rmd file.Rmd". The location is set to "Desktop". The dialog also shows a list of files and folders on the Desktop, including "teaching folder", "Stat240_New lecture notes", "STAT240", "stat240_grades_finalized.csv", "stat240_grades_updated.csv", "scanned exam", "FA25_240_Final_sahifa", "Overleaf backup", "STAT240", "For research", "paper_1_MDPI_submission__revised_10-30", and "New Folder". The "Save" button is highlighted.

Let's understand the different parts of an .Rmd file



The screenshot shows the RStudio editor interface with a file named 'Source'. The editor is divided into two main sections: 'Source' and 'Visual'. The 'Source' section contains the R Markdown code, and the 'Visual' section shows the rendered output. The code is as follows:

```
1 ---
2 title: "Demo rmd file"
3 author: "Sahifa"
4 date: "2026-01-03"
5 output: html_document
6 ---
7
8 ```{r setup, include=FALSE}
9 knitr::opts_chunk$set(echo = TRUE)
10 ```
11
12 ## R Markdown
13
14 This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS
15 Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.
16
17 When you click the Knit button a document will be generated that includes both content as well as
18 the output of any embedded R code chunks within the document. You can embed an R code chunk like this:
19
20 ```{r cars}
21 summary(cars)
22 ```
23
```

Annotations in the image point to specific parts of the code:

- YAML**: Points to the YAML front matter (lines 1-6).
- Global R chunk**: Points to the R code chunk (lines 8-10).
- Body**: Points to the main body of the document (lines 12-16).
- R chunk**: Points to the R code chunk (lines 18-22).



```
1 ---
2 title: "Demo rmd file"
3 author: "Sahifa"
4 date: "2026-01-03"
5 output: html_document
6 ---
```

YAML

YAML: This is usually present in the beginning of an R Markdown file and contains the command for the output file. Such as the title of the document, authorship, date and format. It is defined in between pairs of three dashes '-'. This part is mandatory for every R Markdown file.

```
18 ```{r cars}
19 summary(cars) # this creates a summary of the cars dataset
20 ```
```

R chunk

- R chunk: This is where all the R code goes in an R Markdown. Any R code outside of this gray box is not considered code by the R Markdown file and will not run or appear in the output file. It is defined with three backticks (`) and {r}, and ends with three backticks (`). R chunks can be executed/run by clicking the green triangle on the right or by using 'Ctrl + Enter' for Windows and 'command + Enter' for MAC.
- A new R chunk can be inserted using the shortcut 'Ctrl + Alt + I' for Windows and 'command + option + I' for MAC or by clicking the  on the top right of the documents tab and selecting R from the drop-down menu.
- Any commentary needed in the R chunk can be added by inserting # in front of the text.



```
8 {r setup, include=FALSE}
9 knitr::opts_chunk$set(echo = TRUE)
10 {r }
```

Global R chunk

Global R chunk: This is where a general rule for all R chunk is provided for the output file. For example, 'echo = TRUE' makes all the R chunks appear in the output unless set otherwise, 'error = TRUE' overrides the R Markdown file's default setting to stop at the first error, and also makes the R chunks with errors appear in the output, and so on. Every R chunk in the output file will follow this chunk unless specified otherwise for any individual chunk.

```
11
12 ## R Markdown
13
14 This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS
Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.
15
16 When you click the Knit button a document will be generated that includes both content as well as
the output of any embedded R code chunks within the document. You can embed an R code chunk like this:
```

Body

Body: Any text entered anywhere else on the R Markdown file serves as the body of the output document. Headings are created with hashtags (#), with more hashtags creating subheadings. These also automatically create sections and subsections in the document's layout. Regular text creates the body of the paragraph. Bold text can be created using two asterisks (*) on both sides of the text, whereas italic text needs one asterisk (*) on both sides of the text. Bullet points can be created by putting a dash (-) or numbering in front of the text. Text encased in backticks appear in a gray box in the output. External links can be inserted as [text](hyperlink). An example is shown on the next slide.



Source .Rmd file

Output

```
20 - ## 1) Vector operations
21
22 Remember that in R, if an operation works on a single number, it will
23 usually also work entry-wise on a vector. For example, if you multiply a
24 number by a vector, each entry in the vector will be multiplied by that
25 number. If you multiply two vectors of the same length, the first number
26 of both vectors will be multiplied, and the second number of both
27 vectors will be multiplied, etc. A similar pattern will also work for
28 functions like exp() or pnorm(). Applying a function like these to a
29 vector will apply the function to every entry of the vector. Functions
30 that do this are called *vectorized*.
31
32 a. Create a vector of the numbers 1 to 25 (try to do this without
33 writing out each individual number). Multiply the vector by 2 to get
34 a vector of all the even numbers less than or equal to 50.
35 b. Next, square this vector. Check that the entries of the result are
36 the squares of the first 25 positive even numbers.
37 c. Find the mean of this vector and subtract it from each element of
38 the vector.
39 d. Using >=, compare this vector with 0 to show if each number is
40 greater than or equal to 0. Use sum() on this resultant vector to
41 count how many numbers satisfy this criterion. Alternatively, use
42 mean() to get the proportion (think about why this works!).
43 e. Divide the interval  $(0,1)$  into 15 evenly spaced numbers (**not
44 including 0 and 1). (Hint: see ?ppoints). Then, use qnorm() to
45 get a vector of 15 points evenly spaced out along the quantiles of
46 the normal distribution. Note: this is how you obtain the
47 theoreticals for a
48 QQ-plot (https://en.wikipedia.org/wiki/Q%E2%80%93plot), which you
49 may have seen in your intro classes.
50
```

1) Vector operations

Remember that in R, if an operation works on a single number, it will usually also work entry-wise on a vector. For example, if you multiply a number by a vector, each entry in the vector will be multiplied by that number. If you multiply two vectors of the same length, the first number of both vectors will be multiplied, and the second number of both vectors will be multiplied, etc. A similar pattern will also work for functions like `exp()` or `pnorm()`. Applying a function like these to a vector will apply the function to every entry of the vector. Functions that do this are called *vectorized*.

- Create a vector of the numbers 1 to 25 (try to do this without writing out each individual number). Multiply the vector by 2 to get a vector of all the even numbers less than or equal to 50.
- Next, square this vector. Check that the entries of the result are the squares of the first 25 positive even numbers.
- Find the mean of this vector and subtract it from each element of the vector.
- Using `>=`, compare this vector with 0 to show if each number is greater than or equal to 0. Use `sum()` on this resultant vector to count how many numbers satisfy this criterion. Alternatively, use `mean()` to get the proportion (think about why this works!).
- Divide the interval $(0, 1)$ into 15 evenly spaced numbers (**not including** 0 and 1). (Hint: see `?ppoints`). Then, use `qnorm()` to get a vector of 15 points evenly spaced out along the quantiles of the normal distribution. **Note:** this is how you obtain the theoreticals for a QQ-plot, which you may have seen in your intro classes.



Mathematical equations can also be inserted in R Markdown files by using LaTeX syntax and encasing the LaTeX text in dollar signs (\$). Double dollar signs create **display math mode** (equations on their own, centered).

Source .Rmd file:

1. Given an ``n`` and ``k``, compute the binomial coefficient,

```
$$\binom{n}{k} = \frac{n!}{(n-k)!k!}.$$ You may assume that `n` and `k` are both integers and that $0 \le k \le n$. You may find the `factorial` function in R helpful. See `?factorial` for more information. Be careful that your function correctly handles cases like when $n=0$.
```

Output

1. Given an `n` and `k`, compute the binomial coefficient,

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}.$$

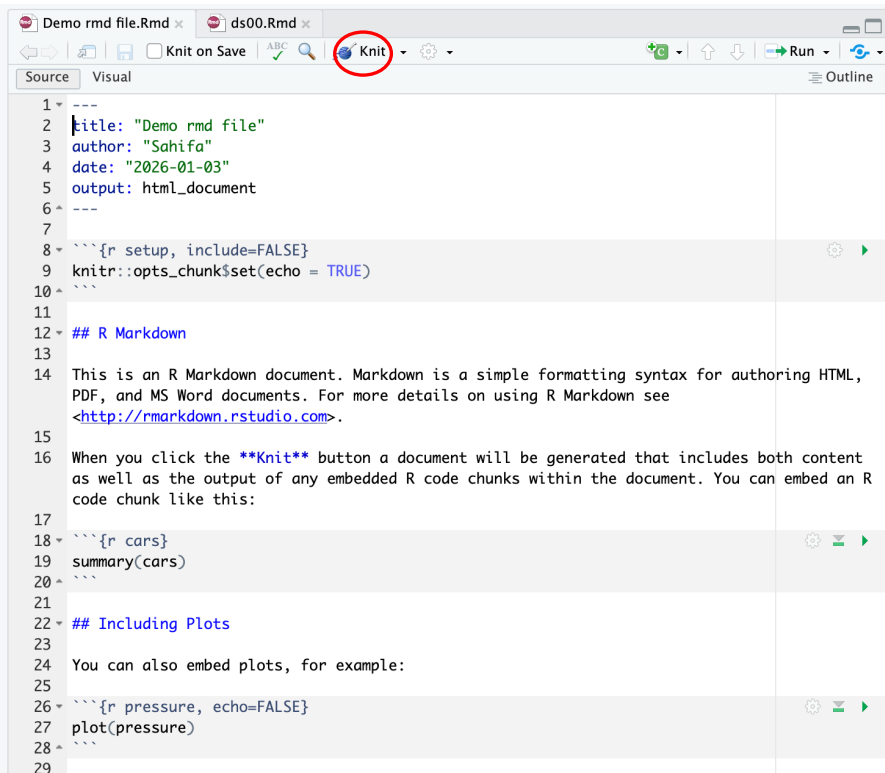
You may assume that `n` and `k` are both integers and that $0 \leq k \leq n$. You may find the `factorial` function in R helpful. See `?factorial` for more information. Be careful that your function correctly handles cases like when $n = 0$.



How is the output file created?

Up until now, we have only seen how to manage or organize an R Markdown file. But how is the output file created? What does it look like?

To create the output file, mostly the html file, we need to `knit` the R Markdown file. This can be done by clicking on the  Knit icon in the top center of the document tab on the source pane. The output html file generated will be saved in the same folder as the original .Rmd file.



```
1 ---
2 title: "Demo rmd file"
3 author: "Sahifa"
4 date: "2026-01-03"
5 output: html_document
6 ---
7
8 ```{r setup, include=FALSE}
9 knitr::opts_chunk$set(echo = TRUE)
10 ```
11
12 ## R Markdown
13
14 This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML,
15 PDF, and MS Word documents. For more details on using R Markdown see
16 <http://rmarkdown.rstudio.com>.
17
18 When you click the Knit button a document will be generated that includes both content
19 as well as the output of any embedded R code chunks within the document. You can embed an R
20 code chunk like this:
21
22 ```{r cars}
23 summary(cars)
24 ```
25
26 ## Including Plots
27
28 You can also embed plots, for example:
29
30 ```{r pressure, echo=FALSE}
31 plot(pressure)
32 ```
```



Demo rmd file

Sahifa

2026-01-03

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

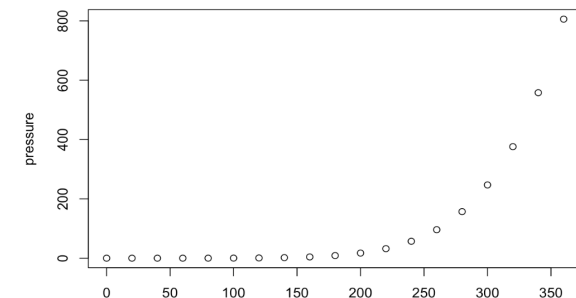
When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

```
##      speed      dist
##  Min.   : 4.0   Min.   : 2.00
##  1st Qu.:12.0   1st Qu.: 26.00
##  Median :15.0   Median : 36.00
##  Mean   :15.4   Mean   : 42.98
##  3rd Qu.:19.0   3rd Qu.: 56.00
##  Max.   :25.0   Max.   :120.00
```

Including Plots

You can also embed plots, for example:





Why an R Markdown file?

- RStudio provides other file formats such as the R script (.R file) but it can only save the code. R scripts are ideal for writing and executing code for analysis. Any text inserted has to begin with hashtags and cannot be anything other than regular text.
- R Markdown is preferred over other R format files due to its ability to create a single, dynamic document that combines code, commentary, and results into a high-quality report.
- Its outputs are often compatible with regular computer software, making it easier to view the documents. Since it also includes the output of the codes, it makes it easier to share your work with others.

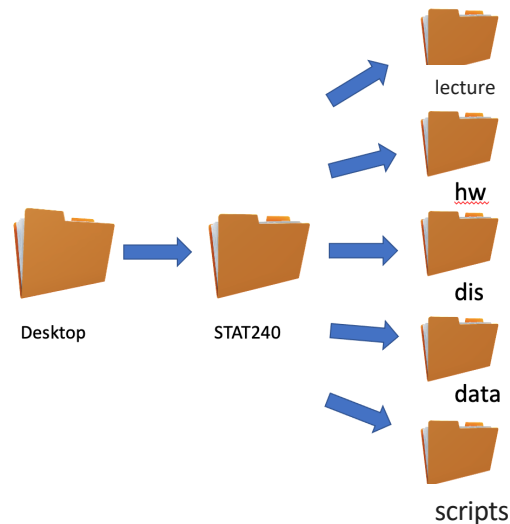
Useful Link: <https://bookdown.org/yihui/rmarkdown/rmdformats.html>



Setting the Directory for STAT 240 in your devices

- Your HWs and DIS will use these paths that will help automatically load datasets and save your files.
- You can do it manually, but it's a little complicated. You can write your own paths, which I will show you how to do later, but I don't recommend it until and unless you know what you are doing.

All of you should set up your directories to look like this:





All of your lecture RMD files, HWs, and DIS will already have the path to the directories written for you. You don't need to change it if you set your directories the way shown in the previous slides.

If needed, you can manually write the path to directories:

Mac Users: “~/Desktop/STAT 240/name of the folder that you want to go to/name of the next folder/.../name of the file”

Window Users: “C:\User\Desktop\STAT 240\name of the folder that you want to go to\name of the next folder\.../name of the file”



Setting working directory in RStudio

Setting the working directory in R is crucial because it establishes a base location for all file input and output operations, ensuring that the script can reliably find source files and save output in a predictable place. There are several ways to do this. Two of the main ways you can use for this course:

- Using the console (the most effective method):

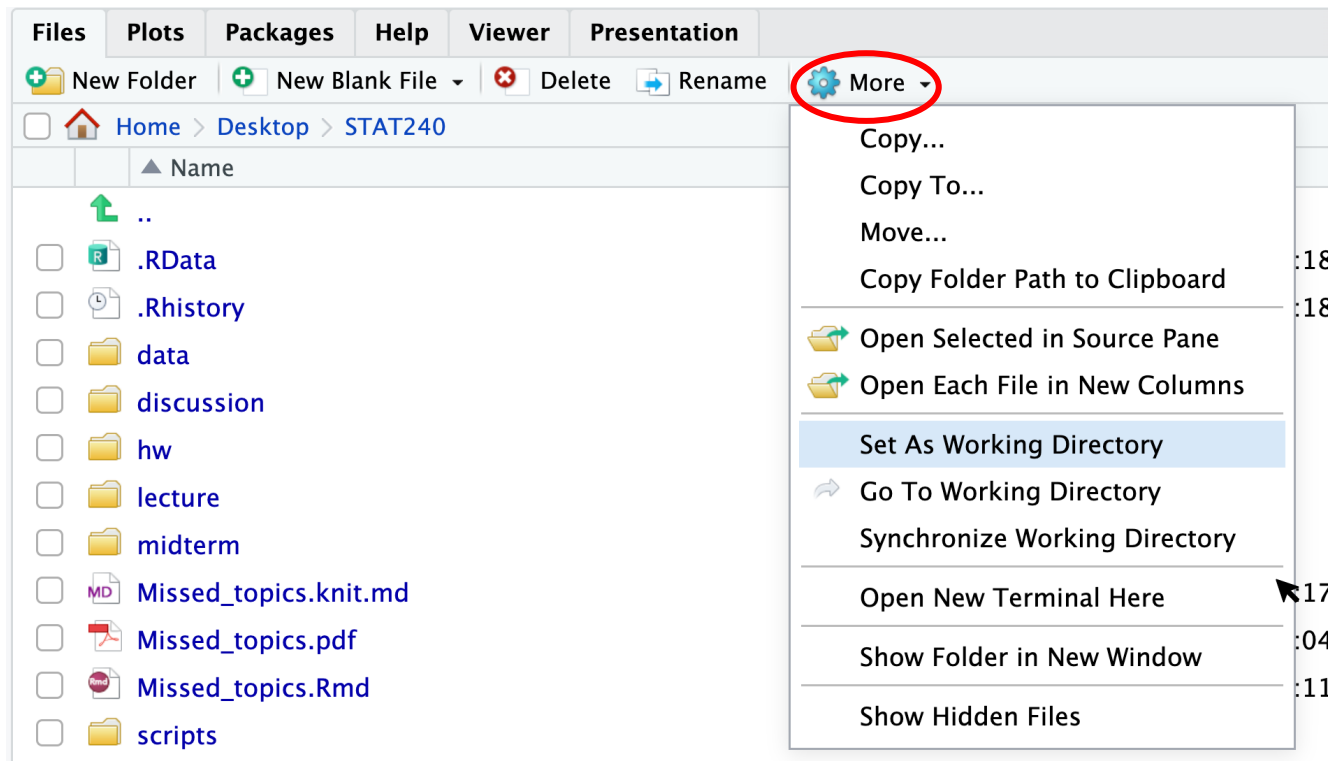
You can use the code `setwd("path to the folder")` to set a certain folder as the working directory. You can get an idea what your working directory is by using ``getwd()`` in the console. For example, I will set the STAT240 folder as my working directory:

```
> getwd()
[1] "/Users/s.sahifa98"
> setwd("~/Desktop/STAT240")
> getwd()
[1] "/Users/s.sahifa98/Desktop/STAT240"
```



- Using the Files tab (the shortcut):

You can navigate to the desired folder in the Files tab, then use the widget's drop-down menu to set it as the working directory by clicking “Set As Working Directory”. You can check where the working directory is by clicking “Go To Working Directory”.



This method poses problems for some Windows user as it's difficult to locate the Desktop in the Files tab, you can use method in the previous slide or come meet me.



We end these lecture slides here. The next lecture will cover the basic syntax of R and different types of objects in R.

